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Urethral implant

Abstract

A urethral-implant is configured to be implanted within a restricted location of a urethra. The urethral-implant forms at least one closed-shape by coiling ends of a single wire section on each other. The closed-shape includes a distal section, a proximal section and two lateral sections extending between the proximal section and the distal section. The closed-shape exhibits an expanded configuration and is foldable into a compressed configuration. The urethral-implant further is configured to extend from the compressed configuration to the expanded configuration so that the distance between wire sections increases, relative to the distance between wire sections in the compressed configuration. The urethral-implant applies continuous pressure on surrounding tissue thereby applying continuous radial pressure on at least one of a urethral wall and tissue surrounding the urethral wall. The radial pressure is sufficient to cause either a widening effect, extending a urinal passage or inducing infarction.

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Background/Summary

(1) This application is a National Stage application of PCT/IL2019/05727, filed Jul. 1, 2019, which claims priority to U.S. Provisional Patent Application No. 62/692,792 filed Jul. 1, 2018 which applications are incorporated herein by reference. To the extent appropriate, a claim of priority is made to each of the above-disclosed applications.

FIELD OF THE DISCLOSED TECHNIQUE

(1) The disclosed technique relates to implants, in general, and to urethral implants produced from a single wire section, in particular.

BACKGROUND OF THE DISCLOSED TECHNIQUE

(2) The urethra is the tube that carries the urine from the bladder to the external urethral orifice, to remove the urine from the body. In some cases, the urethra may be partially or fully blocked due to conditions such as prostate enlargement or urethral stricture, resulting in conditions such as Benign Prostatic Hyperplasia (BPH), prostatic Bladder Neck Obstruction (BNO) and the like, and the flow of fluids in the urethra is restricted and even blocked. Such a partial or full block of the urethra may cause discomfort and pain and may even result in infections.

(3) Transurethral incision of the prostate (TUIP) is an endoscopic procedure usually performed under general anaesthetic in which a surgeon employs an instrument (e.g., a scalpel, a laser beam generator and an electrical current actuator) inserted into the urethra for making incisions in the bladder neck where the prostate meets the bladder (i.e., more specifically in the midline to the level of the verumontanum). Incising the muscles in the bladder neck area relieves the obstructive effect of the prostate on the bladder neck and prostatic urethra and relaxes the opening of the bladder, thus decreasing resistance to the flow of urine out of the bladder. It is noted that, no tissue is removed during TUIP.

(4) Infarction is a process resulting in a macroscopic area of necrotic tissue in some organ caused by loss of adequate blood supply. The inadequate blood supply can result from pressure applied to the blood vessels. Even by applying a relative small but continuous pressure on a tissue, one can block the tiny blood vessels within the tissue and induce infarction.

(5) Known in the art are implants which are inserted through the urethral orifice, in a compressed state to the blocked section of the urethra. When positioned in the blocked section of the urethra, the implant expands and widens the blocked section thus enabling the flow of fluids through the urethra. PCT Patent Application Publication WO 2006/040767 A1, to Kilemnik, and entitled "Prostate Treatment Stent" is directed at a tissue dissecting implant. The implant is spring-shaped and includes a plurality of rings elastically coupled there-between. Adjacent rings apply pressure on tissues caught between the rings, thereby pinching the caught tissues and inducing necrosis.

(6) US Patent Application Publication No. 2011/0276081 to Kilemnik, and entitled "Radial Cutter Implant" is directed at an implant for applying radial forces on the tissues surrounding it. The implant includes wires for applying radial pressure on the surrounding tissues. Each of the wires extends in a different radial direction, and therefore, each wire applies pressure on different tissues. The implant can further include a longitudinal central tube, such that the wires are coupled with a proximal end and a distal end of the tube. The tube supports the wires and provides structural stability to the implant. The distal end of the wires is positioned within the bladder of the subject, and may irritate the bladder.

(7) U.S. Pat. No. 5,209,725 issued to Roth, and entitled "Prostatic Urethra Dilatation Catheter System and Method", is directed to an instrument for performing a transurethral balloon dilatation procedure of the prostate. The balloon dilatation instrument includes a hollow catheter and optical viewing means. The hollow catheter includes a shaft, an inflatable optically transparent balloon, and at least one suitable visible marking. The distal end portion of the shaft is made of an optically transparent material. The inflatable optically transparent balloon is coupled with the distal end portion of the shaft, and is sized to dilate the prostatic urethra. The at least one suitable visible marking is positioned on the catheter proximally to the balloon, such that the marking can be

visualized relative to a predetermined anatomical landmark (e.g., verumon tanum). In this manner, proper positioning of the balloon, relative to the prostatic urethra, is performed prior to and during the dilation of the prostatic urethra. The optical viewing means is slidable within the catheter, for visibly viewing the marking intra-luminally from within the catheter. The balloon is correctly located relative to the prostatic urethra. The balloon is inflated so as to dilate the prostatic urethra without damaging the external sphincter at the apex of the prostate.

(8) U.S. Pat. No. 5,499,994 issued to Tihon et al., and entitled "Dilation Device for the Urethra", is directed to a dilation device for opening a portion of an obstructed urethra. The dilation device includes an inner hollow tubular core and an outer confining covering. The inner hollow tubular core defines a lumen therein. The lumen is a conduit of sufficient diameter to permit urine to flow freely there-through from the bladder. The core is substantially non-collapsible. The outer confining covering is capable of expanding radially outwardly to a predetermined extent. The covering has a length of at least partially that of the obstructed portion of the urethra. The dilation device can further include retractable spikes for anchoring the device in its intended position.

SUMMARY OF THE PRESENT DISCLOSED TECHNIQUE

(9) A urethral implant configured to be implanted within a restricted location of a urethra. The urethral implant, forms at least one closed shape by coiling ends of the single wire section on each other. The at least one closed shape includes a distal section, a proximal section and two longitudinal sections extending between the proximal section and the distal section. The at least one closed shape exhibits an expanded configuration and is foldable into a compressed configuration. The urethral implant further being configured to extend from the compressed configuration to the expanded configuration such that the distance between sections of the wire increases, relative to the distance between the sections of the wire in the compressed configuration. The urethral implant being for applying continuous pressure on surrounding tissue, thereby applying continuous radial pressure on at least one of a urethral wall and tissue surrounding the urethral wall in the restricted location. The radial pressure being sufficient to cause at least one of, a widening effect, extending a urinal passage, and inducing infarction.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The disclosed technique will be understood and appreciated more fully from the following detailed description taken in conjunction with the drawings in which:
- (2) FIG. 1 is a schematic illustration of a urethral implant, constructed and operative in accordance with an embodiment of the disclosed technique;
- (3) FIG. 2 is a schematic illustration of a urethral implant, constructed and operative in accordance with another embodiment of the disclosed technique;
- (4) FIG. 3 is a schematic illustration of a urethral implant, constructed and operative in accordance with a further embodiment of the disclosed technique;
- (5) FIG. 4 is a schematic illustration of a urethral implant, constructed and operative in accordance with another embodiment of the disclosed technique;
- (6) FIGS. 5A-5C are schematic illustrations of deployment of a urethral, operative in accordance with a further embodiment of the disclosed technique;
- (7) FIGS. 6A-6E are schematic illustrations of a urethral implant, constructed and operative in accordance with another embodiment of the disclosed technique; and
- (8) FIGS. 7A-7C are schematic illustrations of a urethral implant, constructed and operative in accordance with another embodiment of the disclosed technique.

DETAILED DESCRIPTION OF THE EMBODIMENTS

- (9) The disclosed technique overcomes the disadvantages of the prior art by providing a urethral

implant for applying continuous pressure on the surrounding tissue, thereby causing ischemia, which results in necrosis. The urethral implant is made of a single wire section, forming at least one closed shape which includes a distal section, a proximal section and two longitudinal sections extending between the proximal section and the distal section. The closed shape exhibits an expanded configuration and is foldable into a compressed configuration. The urethral implant is further configured to extend from the compressed configuration to the expanded configuration such that the distance between sections of said wire increases, relative to the distance between these sections in the compressed configuration. The urethral is configured to be implanted within urethra. For example, the urethral implant may be implanted at a urethral stricture location, for applying pressure on the surrounding tissue (i.e., the inner wall) of the urethra, thereby causing ischemia at the stricture location, which results in necrosis and the widening of the urethra, allowing fluid to flow through the urethra. The urethral implant may include at least two closed shapes where each closed shape has a proximal section, a distal section and two longitudinal sections extending between the respective proximal section and the distal section. Adjacent pairs of the at least two closed shapes share at least one common longitudinal section.

(10) Reference is now made to FIG. 1, which is a schematic illustration of a urethral implant, generally referenced **100**, constructed and operative in accordance with an embodiment of the disclosed technique. Urethral implant **100** is made of a single wire section **102**, forming a closed shape, which includes a distal section **104**, a proximal section **106** and two longitudinal sections **1081** and **1082** extending between distal section **104** and the proximal section **106**. Urethral implant **100** is formed by creating the closed shape with wire **102** and then coiling wire **102** on itself.

(11) Reference is now made to FIG. 2, which is a schematic illustration of a urethral implant, generally referenced **150**, constructed and operative in accordance with another embodiment of the disclosed technique. Urethral implant **150** is made of a single wire section **152**, forming two closed shapes. The first closed shape includes a distal section **154.sub.1**, a proximal section **156.sub.1** and two longitudinal sections **158.sub.1** and **158.sub.2** extending between distal section **154.sub.1** and the proximal section **156.sub.1**. The second closed shape includes a distal section **154.sub.2**, a proximal section **156.sub.2** and two lateral longitudinal sections **158.sub.2** and **158.sub.3** extending between distal section **154.sub.2** and the proximal section **156.sub.2**. Lateral Longitudinal section **158.sub.2** is a common lateral section between the two closed shapes. Urethral implant **150** is formed by first creating the first closed shape with wire **152** and then coiling wire **152** on itself. Coiling wire **152** on itself creates two wire end extensions of wires section **152**. Then, the second closed shape is formed from the two wire end extensions of wire **152** and coiling the two wire end extensions of wire **152** one with the other.

(12) Reference is now made to FIG. 3, which is a schematic illustration of a urethral implant, generally referenced **200**, constructed and operative in accordance with a further embodiment of the disclosed technique. Urethral implant **200** is made of a single wire **202**, forming three closed shapes. The first closed shape includes a distal section **204.sub.1**, a proximal section **206.sub.1** and two longitudinal sections **208.sub.1** and **208.sub.2** extending between distal section **204.sub.1** and the proximal section **206.sub.1**. The second closed shape includes a distal section **204.sub.2**, a proximal section **206.sub.2** and two lateral longitudinal sections **208.sub.2** and **208.sub.3** extending between distal section **204.sub.2** and the proximal section **206.sub.2**. The third closed shape includes a distal section **204.sub.3**, a proximal section **206.sub.3** and two lateral longitudinal sections **208.sub.3** and **208.sub.1** extending between distal section **204.sub.3** and the proximal section **206.sub.3**. Thus, lateral longitudinal section **208.sub.1** is common to the first and the third closed shapes. Thus, lateral longitudinal section **208.sub.2** is common to the first and the second closed shapes and lateral longitudinal section **208.sub.2** **208.sub.3** is common to the second and the third closed shapes.

(13) Urethral implant **200** is formed by first creating the first closed shape wire **202** and then

coiling wire **202** on itself forming longitudinal sections **208.sub.1** and **208.sub.2** similar to as described above in conjunction with FIG. **1**. Then, the second closed shape is formed with the wire end extensions of wire **202** and coiling the two extensions of wire **202** one with the other, similar to as described above in conjunction with FIG. **2**, thus forming longitudinal section **208.sub.3**. The third closed shape is formed with the two wire end extensions of wire **202**, by coiling the two wire end extension of wire **202** on longitudinal section **208.sub.1**.

(14) Reference is now made to FIG. **4**, which is a schematic illustration of a urethral implant, generally referenced **250**, constructed and operative in accordance with another embodiment of the disclosed technique. Urethral implant **250** is similar in construction form and functionality to urethral implant **200** (FIG. **3**).

(15) Each of the single wire urethral implants **100**, **150**, **200** and **250** described hereinabove is made from materials such as stainless steel or Shape Memory Alloy (SMA), such as Nickel Titanium alloy (Nitinol). The transition between the compressed configuration to the expanded configuration occurs due to the elastic (i.e., spring like) characteristics of the material. Each of the single wire urethral implants **100**, **150**, **200** and **250** described hereinabove may be wider at the proximal end than at the distal end thereof for preventing said removable implant from moving in the distal direction. Alternatively, each of the single wire urethral implants **100**, **150**, **200** and **250** described hereinabove may be wider at the distal end than at the proximal end thereof for preventing said removable implant from moving in the proximal direction. According to yet another alternative, each of the single wire urethral implants **100**, **150**, **200** and **250** described hereinabove may be narrower at the middle thereof than at the distal and proximal portions thereof, for preventing said removable implant from moving in the proximal direction and in the distal direction. According to yet another alternative, each of the single wire urethral implants **100**, **150**, **200** and **250** described hereinabove may exhibit the shape of a butterfly wing.

(16) Reference is now made to FIGS. **5A-5C** which are schematic illustrations of deployment of a urethral implant **300**, operative in accordance with a further embodiment of the disclosed technique. Initially, urethral implant is in a compressed configuration and is located within an overtube **302** also referred to as a restrainer. Overtube **302**, with urethral implant **300** compressed therein is inserted into the urethra **304**, until urethral implant **300** is aligned (FIG. **5B**) with a stricture **306**. Thereafter, overtube **302** is pulled over urethral implant **300**. Once overtube **302** is removed from over urethral implant **300**, urethral implant **300** expands to the expanded configuration thereof (FIG. **5C**). As urethral implant extends from the compressed configuration to the expanded configuration, urethral implant **300** applies a force on stricture **306**. As a result, urethral implant **300** applies pressure on the necrosis or scar tissue at stricture **306** thereby dissecting the necrosis or scar tissue and widening urethra **304** at the stricture location. After the treatment, the implant may be pulled out of urethra **304** by reinserting overtube **302** and pulling on lead wire **308** to reinsert urethral **304** into over tube **302**. Over tube **302** is then pulled out of urethra. Alternatively, urethral implant **300** may be pulled out of urethra **304** directly by pulling on lead wire **308** (i.e., without reinserting overtube **302**).

(17) In FIGS. **5A-5C**, urethral implant **300** includes a concave portion in at least one of the longitudinal sections thereof. In the expanded configuration, tissue fills this concave portion, thereby anchoring urethral implant in position.

(18) The exemplary urethral implants described hereinabove are made of a single wire and exhibit a “closed loop configuration”. In a “closed loop configuration”, all the longitudinal sections are connected to both the proximal and the distal sections. However, the urethral implants according to the disclosed technique may also exhibit an “open loop configuration”, where at least one of the longitudinal sections is connected only to the proximal section or only to the distal section. Also, the urethral implants according to the disclosed technique may be cut out of a metal sheet or tube.

(19) Reference is now made to FIGS. **6A-6E**, which are schematic illustrations of a urethral implant, generally referenced **320**, constructed and operative in accordance with another

embodiment of the disclosed technique. Urethral implant **320** includes three longitudinal sections **322**, **324** and **326** arranged in a closed loop configuration (i.e., longitudinal sections **322**, **324** and **326** are connected to proximal section and the distal section of urethral implant **320**). Urethral implant **320** is cut out from a metal sheet or tube. FIG. **6A** depicts a top view of urethral implant **320** in a compressed configuration. FIG. **6B**, depicts cross-section B-B of urethral implant **320** where longitudinal sections **322**, **324** and **326** (as shown) are coplanar, in the compressed configuration. FIG. **6C** depicts an isometric view of urethral implant **320** in a compressed configuration. FIG. **6D** depicts an isometric view of urethral implant **320** in an expanded configuration. FIG. **6E**, depicts cross-section A-A of urethral implant **320** in the expanded configuration.

(20) Reference is now made to FIGS. **7A-7C**, which are schematic illustrations of a urethral implant, generally referenced **350**, constructed and operative in accordance with a further embodiment of the disclosed technique. Urethral implant **350** includes three longitudinal sections **352**, **354** and **356** arranged in an open loop configuration (i.e., where at least one of the longitudinal sections is connected only to the proximal section or only to the distal section). Urethral implant **350**, includes longitudinal section **356** that is connected only to the proximal section. Urethral implant **350** is also cut out from a metal sheet or tube. FIG. **7A** depicts a top view of urethral implant **350** in a compressed configuration. FIG. **7B** depicts an isometric view of urethral implant **350** in an expanded configuration. FIG. **7C** depicts a side view of urethral implant **350** in the expanded configuration.

(21) It will be appreciated by persons skilled in the art that the disclosed technique is not limited to what has been particularly shown and described hereinabove. Rather the scope of the disclosed technique is defined only by the claims, which follow.

Claims

1. A urethral implant configured for implantation within a restricted location of a urethra, said urethral implant is made of a sheet material having at least one cutout, and formed from three longitudinal sections that extend between a distal section and a proximal section of said urethral implant, said urethral implant is flexible between: a compressed configuration that enables deployment of said urethral implant to said restricted location; and an expanded configuration that enables said urethral implant to apply continuous pressure on said restricted location, thereby extending said restricted location of said urethra, wherein said three longitudinal sections of said urethral implant are coplanar in said compressed configuration, and non-coplanar in said expanded configuration.
 2. The urethral implant according to claim 1, wherein at least one longitudinal section of said three longitudinal sections is connected to one of said distal section or said proximal section.
 3. The urethral implant according to claim 1, wherein said urethral implant is cutout from a metal material.
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